Data Engineering Capstone Topic Approval Form

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Capstone Project Name: Multiple Linear Regression on X Restaurant Sales Dataset

Project Topic: Predictive model for Restaurant Data

☒ **This project does not involve human subjects research and is exempt from WGU IRB review.**

Research Question: Can multiple linear regression model be constructed on the research dataset?

Hypothesis: **H0**- A predictive MLR model cannot be made from the research dataset
H1- A predictive MLR model can be constructed from the research dataset at a model accuracy > 70%

Context: This study contributes to the field of Data Analytics and the MSDA program by developing a predictive model to estimate restaurant sales and improve inventory planning through understanding environmental influences. A general linear model will evaluate the significance of predictor variables—weather, location, and population—to identify those that best forecast the sales revenue of X Restaurant. In recent years, data analytics in the restaurant and food-service industry has expanded, with studies examining how predictor variables such as daily weather, which affects customer traffic, relate to sales. One investigation found that "to aid in forecasting, we engineer many features and demonstrate good methods to select an optimal sub-set of highly correlated features" (MDPI., 2022), where a linear model achieved a one-day forecasting error (sMAPE) of only 19.6% on gross sales data (MDPI., 2022). Another study highlights that "accurate demand forecasting is one of the key aspects for successfully managing restaurants … properly predicting future sales … is critical to guarantee the profitability of the restaurant" (OUCI., 2022). The authors demonstrated that generalized additive models capture strong seasonal and trend changes in restaurant sales records (OUCI., 2022). Similarly, research on machine-learning-based forecasting in food-service outlets showed that including external factors such as weather, local events, holidays, and promotions can substantially improve accuracy: "the AI model … reduces the root mean squared error by 22% to 33% and the mean average error by 19% to 31%" (Springer., 2024). These findings illustrate how statistical and regression-based models can express relationships between predictor variables and sales through coefficients, significance levels, and model-fit metrics. Accordingly, this study aims to build a predictive model that incorporates environmental and demographic factors—specifically weather and population—to estimate restaurant sales in dollar amounts, enhancing inventory planning and data-driven decision-making.

Data: The data required for this study consists of daily sales, date, and store-city location provided by a private business (the "X Restaurant" dataset). These fields will be supplemented by historical weather data (e.g., temperature, precipitation, humidity) retrieved via the Meteostat Python Library / API interface (Meteostat., 2025). Additionally, city-level estimated population statistics will be incorporated to capture a demand/market-size context variable with an overall table rows of 8,830.

The primary dataset includes a series of daily records for the restaurant (one row per day per city store) containing the date of sales, total dollar amount of sales, and the store's city (or location) identifier. After initial receipt of the core dataset, a data-engineering transformation step adds: the daily weather variables (e.g., average temperature, precipitation), and the estimated population for the city on that date (or most recent available). The merged dataset results in a multi-variable table where the response variable is the daily sales dollar amount, and predictor variables include weather features, date/time features, city population, and potentially store-location categorical variables.

Variable	Type
time	Categorical
tavg	Numerical
tmin	Numerical
tmax	Numerical
prcp	Numerical
wspd	Numerical
pres	Numerical
city	Categorical
month	Categorical
month_name	Categorical
season	Categorical
estimated_population	Numerical
daily_sales	Numerical

Link:https://github.com/hiyabfm/D10-Data-Engineering-Capstone/blob/d867ec8639fc6bcb3a3bf434dbf3724a8609e640/washington_x_restaurant_daily_sales_dataset.csv

The core restaurant sales dataset is owned by a individual with a private business, who has granted permission for its use in this academic capstone project without using the business name or any other information. The weather data are retrieved from meteorological open-data sources via the Meteostat API, which offers free historical weather and climate data (Meteostat., 2025). Because the dataset is non-human-subject (no personally identifiable customer data) and only involves aggregated sales results for the business, the project remains exempt from human-subjects IRB review per your institution's guidelines.

Data Gathering:. The restaurant sales dataset was obtained directly from the owner of a private business and serves as the primary data source for this study. The dataset includes 8,830 daily records containing the date, total sales amount, and store city location. The data cover the period from January 1, 2023, through June 1, 2025, with a total accumulated sales amount of $8,042,880.58. Additional data were collected from the Meteostat API to provide weather-related predictor variables. Using the city's geographic coordinates or the nearest available weather station, daily values for average, maximum, and minimum temperature, precipitation, and humidity were retrieved. The dataset was further enhanced by incorporating estimated city-population figures obtained from publicly available census and demographic data sources to account for variations in local demand. The final dataset underwent a series of data-quality checks and transformation processes. Missing or incomplete records were removed or imputed, date formats were standardized, and categorical encodings were applied for features such as day-of-week and city. Outlier detection was performed to identify and correct abnormal sales values. The overall data quality was high, with minimal missing values after cleaning. The newly added variables will help identify which predictors significantly influence daily sales variable.

Data Analytics Tools and Techniques: The primary analytical technique will be a multiple linear regression (MLR) model (a type of general linear model) where the response variable is the daily sales amount, and the predictor variables include weather metrics, city population, day-of-week, store location, and possibly other temporal features (holiday indicator, month). Prior to fitting the MLR,

exploratory data analysis (EDA) will include: Q–Q plots to assess normality of residuals or transformed sales; scatter plots and correlation matrices to inspect relationships between predictors and the response and possibly stepwise selection or regularization to identify the most significant predictors. The use of Q–Q plots ensures the assumptions of linear regression (normality of residuals) are checked. Correlation matrices and scatter plots help assess the linear associations between variables and support the rationale for linear modelling. Model fit will be evaluated via adjusted R^2 and a hold-out test set to assess prediction accuracy (with the aim of > 70% accuracy under the alternative hypothesis). If any strong non-linear relationships emerge, transformation of predictors or interaction terms may be considered.

Justification of Tools/Techniques: Python will be utilized as the primary programming language for developing the multiple linear regression model in this study. According to Scholar.SMU (2018), "Python is an open-source, general-purpose tool with applications for web, Internet, and software development; education and academia; numeric and scientific," making it one of the most versatile and optimal tools for data engineering and data science tasks. Although alternative platforms such as SAS and R were considered, "SAS does not have the large user-written package library common to R and Python," and R's "main target audience is any user interested in statistical computing" (Scholar.SMU, 2018). Python's extensive libraries, including *pandas*, *scikit-learn*, and *statsmodels*, provide a robust framework for building, testing, and validating predictive models. This model implementation supports hypothesis testing for all variables within the dataset, ensuring a rigorous and interpretable analysis aligned with the study's research objectives.

Project Outcomes: This project aims to develop a multiple linear regression model to predict the daily sales (dollar amount) of X Restaurant based on the combined influence of weather and population variables. Support for the alternative hypothesis is reflected in previous research by OUCI (2022) and Springer (2024), which emphasizes that linear modeling and forecasting are essential for effectively managing and predicting future restaurant sales. The model will estimate the target variable by evaluating the strength and significance of each predictor variable in relation to weather patterns and population data, thereby providing actionable insights for improved business planning and inventory management.

Projected Project End Date: 11/15

Sources:

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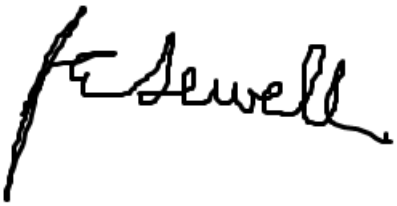
Instructor Signature/Date:

☐ The research is exempt from an IRB Review.

☐ An IRB approval is in place (provide proof in Appendix B).

Instructor's Approval Status: **Approved**

Date: 11/9/2025



Reviewed by:

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